INCH-POUND
MIL-DTL-15291/8C
21 January 2022
SUPERSEDING
MIL-DTL-15291/8B
6 February 2006

DETAIL SPECIFICATION SHEET

# SWITCHES, ROTARY, SNAP ACTION CLASS 6SR FRONT MOUNTED, BACK CONNECTED

Inactive for new design after 17 April 1981. For new design, use MIL-DTL-15291/9

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-DTL-15291.

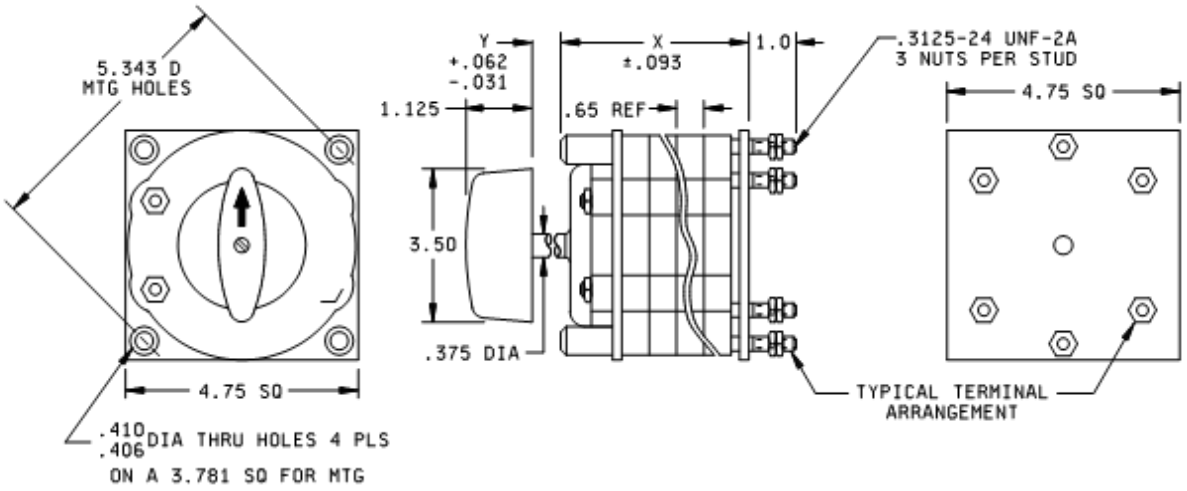



FIGURE 1. Class 6SR switch (up to 6 studs).



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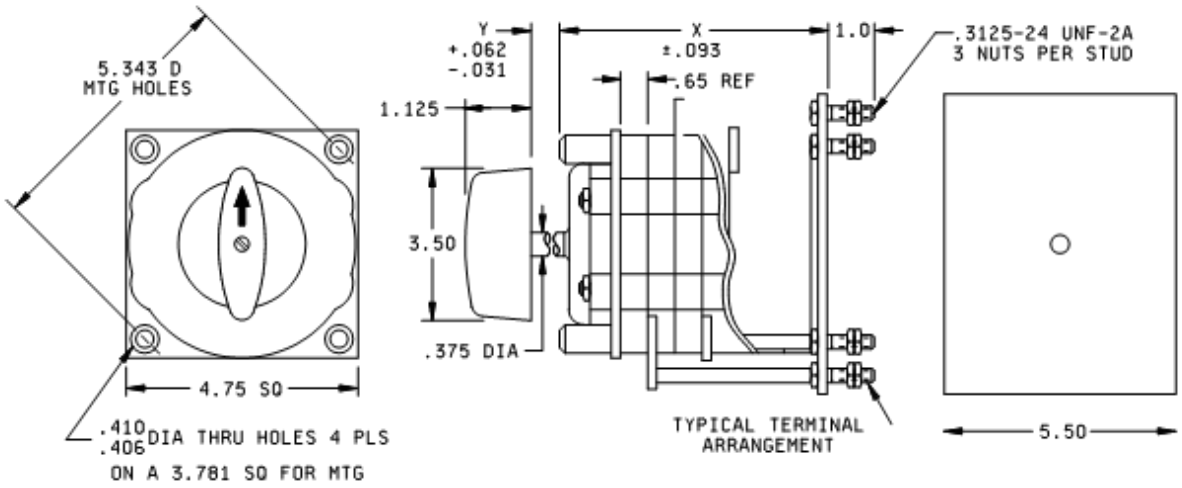
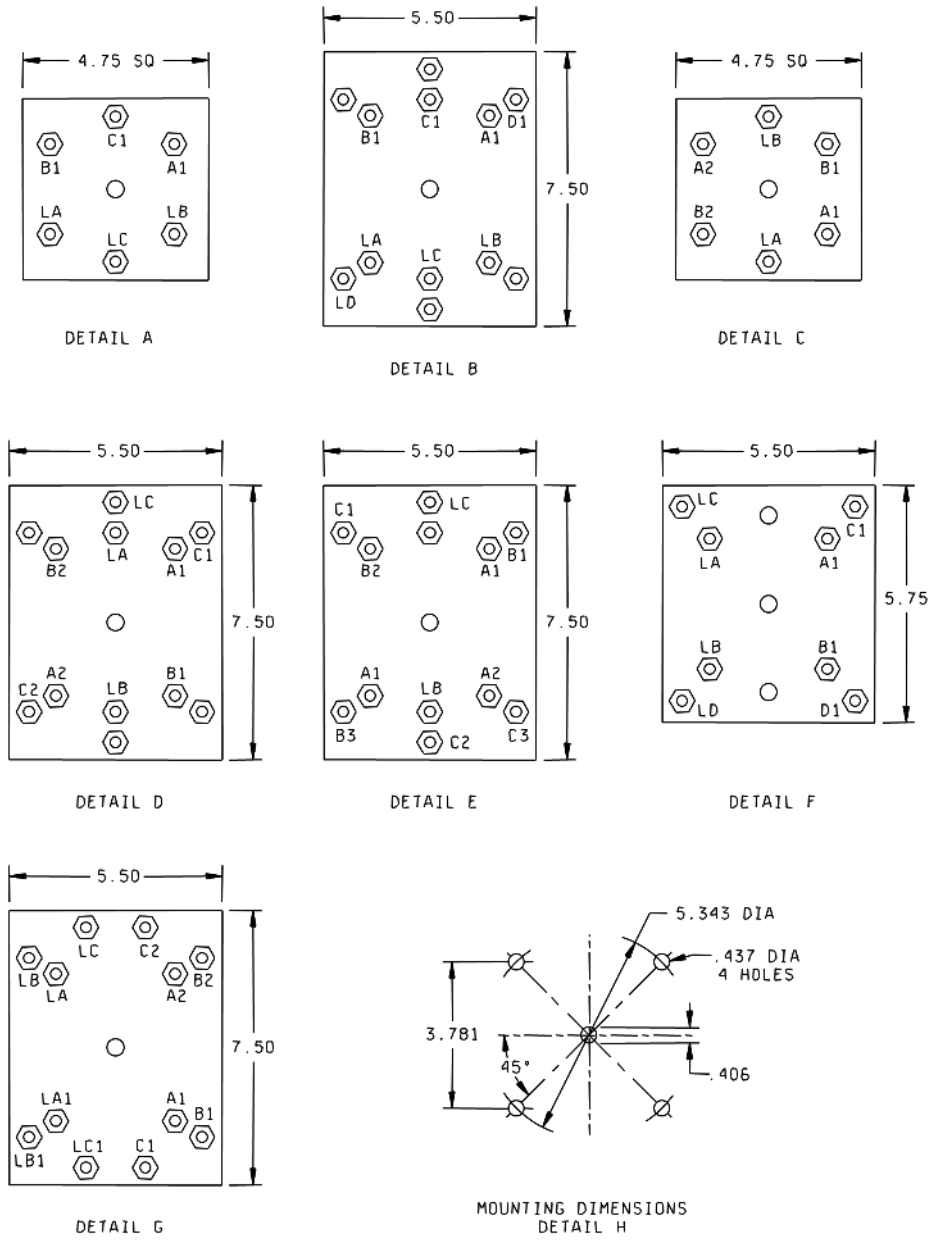



FIGURE 2. Class 6SR switch (up to 12 studs).



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NOTE: Terminal studs viewed from rear.

FIGURE 3. Mounting dimensions and terminal stud locations.



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TABLE I. Type and switching characteristics.

| M15291/8 DASH NO. TYPE DESIGNATION | DETAIL REF | DIM "X" | HANDLE OR SHAFT POSITION | CIRCUIT AND SPACER CONFIGURATIONS ROTOR POSITION, SPACER LOCATIONS, TERMINAL MARKING AND LOCATIONS | | | | | | | | | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | TORQUE IN-LBS | DIM "Y" | | NO 1 | NO 2 | NO 3 | NO 4 | NO 5 | NO 6 | NO 7 | NO 8 | NO 9 | |
| -001 | FIG 1 DETAIL A | 3.437 | OFF O N 1 OFF | | B1 LB | A1 LA | | | | | | | |
| 6SR2A1 | 26 | .437 | | | | | | | | | | | |
| -002 | FIG 1 DETAIL A | 4.093 | OFF O N 1 OFF | | C1 LC | B1 LB | A1 | | | | | | |
| 6SR3A1 | 26 | .437 | | | | | | | | | | | |
| -003 | FIG 2 DETAIL B | 4.750 | OFF O N 1 OFF | | D1 LD | C1 LC | B1 LB | A1 LA | | | | | |
| 6SR4A1 | 26 | .437 | | | | | | | | | | | |
| -004 | FIG 1 DETAIL C | 4.750 | OFF O N 2 1 OFF | | LB | B1 B2 | LA | A2 A1 | | | | | |
| 6SR2B1 | 26 | .437 | | | | | | | | | | | |
| -005 | FIG 2 DETAIL D | 6.062 | OFF O N 2 1 OFF | | LC | C1 C2 | LB | B2 B1 | LA | A1 A2 | | | |
| 6SR3B1 | 26 | .437 | | | | | | | | | | | |
| -006 | FIG 2 DETAIL E | 4.750 | OFF O N 1 3 ON 2 | | B1 LB | B2 B3 | LA | A3 A2 A1 | | | | | |
| 6SR2E1 | 26 | .437 | | | | | | | | | | | |
| -007 | FIG 2 DETAIL E | 6.062 | OFF O N 1 3 ON 2 | | LC | C1 C3 C2 | LB | B1 B2 B3 | LA | A3 A2 A1 | | | |
| 6SR3E1 | 26 | .437 | | | | | | | | | | | |
| -008 | FIG 2 DETAIL F | 4.750 | OFF O N 1 2 1 AND 2 | | D1 LD | C1 LC | B1 LB | A1 LA | | | | | |
| 6SR4F1 | 26 | .437 | | | | | | | | | | | |
| -009 | FIG 2 DETAIL G | 6.062 | OFF O N 1 2 1 AND 2 | | C1 LC1 | C2 LC | B1 LB1 | A1 LA1 | | | | | |
| 6SR6F1 | 26 | .437 | | | | | | | | | | | |



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REQUIREMENTS:

Applicable specification: MIL-DTL-15291.

Dimensions and mounting: See figures 1 through 3.

Switching characteristics: See table I.

Angle of throw: 90 degrees.

Switching action: Snap action, reciprocating.

Electrical and endurance ratings: See table II.

Stop strength (applicable to switches with stops): 100 inch-pounds.

Vibration: 50 Hz, MIL-STD-167-1.

Shock: High impact, MIL-DTL-901.

Contact resistance: 0.005 ohms maximum.

Dielectric withstanding voltage: 2,000 V rms.

Insulation resistance: 200 megohms minimum.

Temperature rise: 50°C maximum.

Mounting bolts: (2), 2/8 diameter, length, and headstyle to suit application. Bolts not furnished.

Handle: M15291/9-100 per MIL-DTL-15291/9.

Terminal marking: Terminal marking shall be in accordance with details A through G, as specified in table I. Terminal markings in table I are shown as viewed from the front (handle end) of the switch.

TABLE II. Electrical and endurance ratings.

| Tests | Current (amperes) | Voltage (volts) | Electrical operations (number of operations) | Test rate (operations per minute) | De-energized operations (number of operations) |
|---|---|---|---|---|---|
| Alternating current (rms) | 60 | 500 | 6,000 | 10 | 4,000 |
| Direct current | 60 | 250 | 4,000 | 5 | --- |



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GENERAL INFORMATION:

Switches not covered by specification sheets: Switches which are fabricated from standard parts, as used in qualified switches, but which do not comply with switches detailed herein with respect to circuit characteristics, switching action, mounting arrangements, and handle details may be acquired under this specification from contractors having qualification approval under this specification.

Extended ratings: Switches detailed herein have been tested and found satisfactory at the extended ratings listed in table III. Reduced life expectancy must be anticipated for switches used in these increased voltage or current levels. Tests under the conditions of table III are not required for qualification acceptance and they are not repeated routinely as for maintenance of qualification. Supplemental evaluations and tests applicable to particular circuit requirements are recommended.



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TABLE III. Extended ratings.

| Switching characteristics | AC 60 or 400 Hz | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 125 volts | | | | 250 volts | | | | 500 volts | | | |
| | Resistive or lamp load | | Inductive load 0.75 p.f. | | Resistive or lamp load | | Inductive load 0.75 p.f. | | Resistive or lamp load | | Inductive load 0.75 p.f. | |
| | Amp. | Operations | Amp. | Operations | Amp. | Operations | Amp. | Operations | Amp. | Operations | Amp. | Operations |
| All | 60 | 6,000 | 60 | 6,000 | 60 | 6,000 | 60 | 6,000 | 60 | 6,000 | 60 | 6,000 |
| E, F | 75 | 3,000 | 75 | 3,000 | -- | -- | -- | -- | -- | -- | -- | -- |
| A, B 1/ | 75 | 3,000 | 75 | 3,000 | 60 | 6,000 | 60 | 3,000 | 60 | 3,000 | 30 | 3,000 |
| | DC | | | | | | | | | | | |
| | 120 volts | | | | 250 volts | | | | 350 volts | | | |
| | Resistive or Lamp load | | Inductive load[2] | | Resistive or Lamp load | | Inductive load[3] | | Resistive or Lamp load | | Inductive load[2] | |
| | Amp. | Operations | Amp. | Operations | Amp. | Operations | Amp. | Operations | Amp. | Operations | Amp. | Operations |
| All | 60 | 3,000 | 50 | 1,500 | 30 | 3,000 | -- | -- | -- | -- | -- | -- |
| A, B | 60 | 4,000 | 60 | 4,000 | 60 | 4,000 | 50 | 3,000 | 30 | 3,000 | 30 | 3,000 |

1/ Single pole break: All other ratings are based on breaking both sides of the line in accordance with figures 1 and 2.
2/ 0.04 henry for inductive circuit.




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APPLICATION AND ACQUISITION GUIDE: PIN and type designation cross reference shall be as shown in table IV.

TABLE IV. Application and acquisition guide.

| M15291/8 dash number | Type designation | For replacement only In new design, use M15291/9 | Circuit configuration |
|---|---|---|---|
| -001 | 6SR2A1 | -001 | A (off-on-off-on) |
| -002 | 6SR3A1 | -002 | A (off-on-off-off) |
| -003 | 6SR4A1 | -003 | A (off-on-off-on) |
| -004 | 6SR2B1 | -008 | B (on 1-off-on 2-off) |
| -005 | 6SR3B1 | -009 | B (on 1-off-on 2-off) |
| -006 | 6SR2E1 | -012 | E (off-on 1-on 2-on 3) |
| -007 | 6SR3E1 | -013 | E (off-on 1-on 2-on 3) |
| -008 | 6SR4F1 | -016 | F (off-on 1-on 1 & 2-on 2) |
| -009 | 6SR6F1 | -017 | F (off-on 1-on 1 & 2-on 2) |

Referenced Documents:

MIL-DTL-15291
MIL-DTL-15291/9
MIL-DTL-901
MIL-STD-167-1

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Custodians:
Navy - SH
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-2022-017)

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